

THE RATIONALIZABILITY OF SURVEY RESPONSES

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ABSTRACT. We propose the concept of survey rationalizability, that we base on classical item response theory in psychology. Survey rationalizability involves positioning questions on a common scale such that each respondent agrees with questions aligned with their views. We use ideas from classical revealed preference analysis to characterize when survey responses are rationalizable. We then show that these results readily extend to polytomous questions and stochastic responses. Furthermore, we investigate the identification of the models.

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1. INTRODUCTION

Surveys are becoming increasingly popular in economics as tools for gathering data on various economic variables, such as subjective expectations, happiness, contingent valuations, beliefs, and political attitudes. Yet, the rationality foundations and the resulting empirical content of survey responses are not sufficiently understood from an economic point of view. This paper takes a first step in this direction by formalizing classical psychological theories of item response and then studying their empirical content using concepts inspired by traditional revealed preference analysis. Ultimately, this type of exercise is crucial for better assessing the external validity of survey information and for improving the structure of surveys in terms of informativeness and accuracy, aspects that ultimately lead to better decision-making and policies.

We study surveys measuring attitudes, opinions, beliefs, or preferences on a given topic. We begin the analysis with the case of dichotomous (yes/no) and deterministic questions, and later argue that this case provides the foundation for studying more complex surveys. We draw on the classical treatment of item response theory with ideal points in psychology.¹ Accordingly, we model rationalizability by placing survey questions on a common scale, shared by all individuals, and assigning each individual a strictly quasi-concave utility over that scale. Then, each individual endorses exactly the questions whose positions give utility above her threshold, which, by strict quasi-concavity, form an individual-specific interval.²

We show that the study of rationalizability can be built on the basis of an intuitive revelation principle. Consider an individual who endorses questions q_2 and q_3 , but not question q_1 . Given the interval nature of responses, the individual reveals that q_1 must occupy an extreme position in the triple. A minimal rationalizability requirement should be that these direct revelations cannot be contradictory; if there are individuals that reveal two questions as extreme within a triple, it cannot be the case that another individual reveals the third question as also extreme in the triple. We refer to this

¹See Thurstone (1928) and the unfolding theory of Coombs (1964) for early developments, and see Tay and Ng (2018) and Van der Linden (2018) for more recent accounts.

²We note here that our formalization of survey rationalizability can be applied to a wide variety of economic settings that deviate from the traditional framework involving choices among alternatives. The common characteristic of these settings is the absence of rivalry among the alternatives. For instance, in the context of surveys, individuals express their degree of conformity with a statement rather than choosing which of several statements they prefer. We discuss this point in greater detail in Section 3.1.

asymmetry type of property as the Weak Axiom of Revealed Extremeness (WARE). We show that the study of direct revelations is not sufficient for rationalizability. Fortunately, we are able to show how to incorporate indirect revelations, something that can be done in two ways. First, we expand survey data with the inclusion of particular fictitious responses in a way that applying WARE to the expanded survey becomes sufficient. Second, we parallel the standard revealed preference analysis by appropriately extending the revealed extremeness relation, and then imposing asymmetry on the extended relation, a property that we call the Strong Axiom of Revealed Extremeness (SARE). In both cases, we are accounting for the exact indirect revelations contained in the data, leading to our characterization of survey rationalizability in the first part of Theorem 1. Moreover, this characterization also helps to show that testing for rationalizability is computationally tractable, as we show in the second part of the theorem.

We then show that our techniques for the dichotomous and deterministic case prove useful in the study of more general settings. In particular, we consider the cases of polytomous surveys, where questions can be answered using one of multiple levels of endorsement, and probabilistic survey data, which allows for belief expressions or noisy responses. We adjust our notion of survey rationalizability to consider two cases: ordered collections of thresholds (one for each endorsement level) and random thresholds. Theorems 2 and 3 establish the characterizations for the first and second cases, respectively. Both results follow from a simple transformation of these richer datasets into analytically equivalent dichotomous datasets, allowing us to use the techniques developed for the proof of Theorem 1.

We then shift our focus to studying the identification of these models. Since they are essentially ordinal, the key step in the analysis is determining the order of questions and the location of individuals' ideal points. When the survey data is rationalizable, the extremeness revelations are key to learning the order of questions. In Proposition 1, we show that a richness condition guarantees that this order is unique. Regarding the location of individuals, Proposition 2 demonstrates that it is possible to establish bounds for them, with precise information being conveyed by probabilistic surveys, whereas polytomous surveys provide more information than dichotomous surveys. Furthermore, in Proposition 3, we show that all locations can be cardinally and uniquely identified when a specific parametric model is considered.

We close the analysis by studying the assumption on the unidimensional location of questions and individuals. We argue that this assumption is not only relevant because a large number of theoretical settings and empirical applications in psychology, sociology, political science, and economics adhere to it, but it is also critical. This is so because the multidimensional analysis can be addressed in a straightforward manner. Essentially, in a multidimensional model, survey rationalizability loses its empirical content; we show in Theorem 4 that every survey can be rationalized when just two dimensions are considered.

2. RELATED LITERATURE

In economics, the empirical literature using surveys is large and rapidly growing. Here, our focus is on its theoretical and methodological counterpart. Bertrand and Mullainathan (2001) propose an econometric-based framework that accounts for errors in responses, enabling a meaningful interpretation of individuals subject to cognitive biases. Prelec (2004) proposes a scoring method for the elicitation of truthful subjective judgments based on answers that are more common than collectively predicted. Falk, Neuber, and Strack (2021) develop an individual-response model based on imperfect self-knowledge, where individuals' responses depend on a combination of private signals and the population mean. Manski (2004) and Stantcheva (2022) discuss several methodological issues in designing and interpreting survey studies. Benjamin, Guzman, Fleurbaey, Heffetz, and Kimball (2023) propose a methodology to uncover the informational content of self-reported well-being surveys, considering various potential response biases. Liu and Netzer (2023) study the use of response time to enhance the informational content of happiness survey responses, aiming to overcome the identification problems established in Bond and Lang (2019). Our contribution to this literature is to offer a theoretical framework that studies the rationality of survey responses based on classical accounts in psychology. We also show how the ordered structure of questions and individual attitudes is revealed. Finally, our probabilistic models of survey responses enable systematic consideration of various types of errors.

Our results relate to various strands of literature across scientific disciplines, such as the use of sortability and seriation in archaeology, anthropology, biology, computer science, and psychology. The common connection among these fields is the so-called consecutive-ones property (c1p) of a 0-1 matrix. A matrix satisfies c1p if there exists a

permutation of its columns such that the ones in every row are consecutive.³ In essence, for the dichotomous and deterministic case, survey rationalizability can be expressed as a c1p problem, enabling the use of existing results.⁴ Importantly, our analysis, inspired by classical revealed preference techniques, provides a conceptually simple and tractable characterization of survey rationalizability and, in doing so, of c1p. Our proof demonstrates that c1p requires only checking asymmetry in an appropriately defined ternary relation. Additionally, we show how our techniques can be extended to address more complex settings, such as those involving polytomous surveys or stochastic responses.

Our paper can also be related to the study of ternary relations. Huntington and Kline (1917) and Fishburn (1971) represent early treatments of abstract ternary relations, with the purpose of capturing the intermediateness notion on the real line. Most of our results are presented in terms of an extremeness relation, but they can be equivalently presented in terms of an intermediateness relation. Our results contribute to this literature both conceptually and technically. Conceptually, we apply ternary relations to a concrete economic problem and, instead of taking them as given, we derive them from data. Technically, we show which incomplete ternary relations can be extended into the ternary relation of a linear order. We achieve this by combining techniques from the c1p literature and the revealed preference literature.

3. THE RATIONALIZABILITY OF DICHOTOMOUS SURVEYS

Let $\mathcal{Q} = \{1, \dots, q, \dots, Q\}$ be a set of questions related to a specific topic, such as capital punishment or immigration, on which individuals $\mathcal{N} = \{1, \dots, n, \dots, N\}$ are surveyed. We interpret question $q \in \mathcal{Q}$ as a request from individuals to provide a graded evaluation or endorsement of the statement described in the question; in the dichotomous case considered here, these evaluations are yes/no, agree/disagree

³As an illustration of an application of c1p in a different field, consider the statistical archaeology problem discussed in Kendall (1969). The data consists of a set of graves and a set of objects that are either present or absent in these graves. C1p is used to determine whether graves and objects can be arranged on a time scale such that each grave contains only objects that were created and not yet obsolete during the relevant period. See Coombs and Smith (1973) and Hubert (1974) for connections between these techniques and survey responses, and Liiv (2010) for an overview of their applications across different disciplines.

⁴See the discussion following the proof of Theorem 1 for the use and connection of our results with classical c1p results, such as Tucker (1972).

or positive/negative evaluations. Accordingly, a dichotomous survey (D-survey) is a collection of the form $\{Q_n\}_{n \in \mathcal{N}}$, with $Q_n \subseteq \mathcal{Q}$ describing the set of questions that are endorsed by individual n .⁵ Without loss of generality, we assume that all response sets are distinct and that no two questions have the same set of endorsing individuals. We study when a D-survey can be rationalized in the following sense.

D-survey rationalizability. We say that $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable whenever there exist $[\{U_n, \tau_n\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave and $\tau_n, \mu_q \in \mathbb{R}$, such that for every $n \in \mathcal{N}$ and $q \in \mathcal{Q}$, it holds that $q \in Q_n$ if and only if $U_n(\mu_q) \geq \tau_n$.

The notion of rationalizability contains two major assumptions: (i) views are to be located on the real line, and (ii) utility functions are strictly quasi-concave. As for the first assumption, unidimensionality is a standard assumption in psychology, sociology, political science, and many economic settings of interest.⁶ In any case, we extend our analysis to the multidimensional case in Section 6. Regarding the second assumption, once views are unidimensional, we believe this modeling is the most natural in economic terms in the presence of trade-offs, and it captures, using standard economic language, the notion of endorsement-by-proximity described in the psychology literature.⁷ Another important advantage of using strict quasi-concavity is that it provides a flexible framework for analyzing more complex survey types, such as those studied later in the paper. In particular, we can address the rationalizability of Likert-type surveys or of probabilistic surveys by retaining strict quasi-concavity and introducing multiple thresholds or probabilistic thresholds, respectively.

Before continuing, note that the interpretation of the utility function depends on the application at hand. Some surveys contain questions where the utility function can be interpreted in line with classical economic terms, even though the question may be hypothetical and utility may result from the hypothetical contemplation of

⁵Note that by considering the incidence matrix of survey responses, we can view a D-survey as a matrix of 0s and 1s.

⁶In many settings, the order of questions is not explicitly known, and we show how it can be learned from the response data. Section 3.1 provides several examples and, in most of them, the order of questions is not necessarily known.

⁷It can be shown that rationalizability can be defined equivalently as the existence of $[\{\mu_n, \epsilon_n\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$ with μ_n representing the location or ideal view of the individual, such that for every $n \in \mathcal{N}$ and $q \in \mathcal{Q}$, it is $q \in Q_n$ if and only if $d(\mu_n, \mu_q) \leq \epsilon_n$, where d is a distance function.

the question. In these situations, the threshold level may represent some internal expectation. In other types of surveys, the utility function may be interpreted as the internal value derived from supporting a specific statement, with rejected statements being those that entail high costs of departing from personal beliefs or views. The specific interpretation is not crucial to our study; the representation we adopt helps to formalize the idea of survey rationalizability in a language common to economists.

3.1. Examples of the setting. Our formal setting uses the language of attitude surveys on a given topic, which represents its most direct application. These surveys are extensively used in economics to gather information on a variety of topics and, similarly, widely used in other disciplines, such as psychology, to measure personality traits or attitudes. Examples of their use can be seen in Andrich (1988), where individuals are asked to endorse or reject a number of statements on capital punishment, and Grable and Lytton (1999), which surveys the risk attitudes of individuals.⁸ However, our setting is also applicable well beyond the survey interpretation. Here, we provide some economic examples that show the potential applicability of our results. The shared characteristic of these examples is that there is little rivalry between the options, and the issue is whether individuals like/use/consider/accept each of the options.⁹ Moreover, all these cases also share the feature that, in order to address the research question at stake, a common ordering of the options is necessary, and this is not clear ex-ante from their nature.

Activities. Behavioral traits, such as risk aversion, can be analyzed by gathering information on individual participation in various potentially risky activities, such as drug use or extreme sports. Similarly, altruism can be assessed by collecting data on participation in pro-social activities. This approach has a long tradition in psychology; an example is Fromme, Katz, and Rivet (1997), who designed and studied the cognitive

⁸The former includes statements/questions such as: (i) Capital punishment is justified because it does act as a deterrent to crime; (ii) I think capital punishment is necessary but I wish it were not; (iii) Until we find a more civilized way to prevent crime we must have capital punishment. The latter includes questions such as: (i) It's hard for me to pass up on a bargain; (ii) It is more important to be protected from rising consumer prices (inflation) than maintaining the safety of your money from loss and theft; (iii) Are you comfortable investing in stocks or stock mutual funds? Note that these questions do not have a clear ordering.

⁹We use the dichotomous case, but the more general cases with Likert scales and probabilistic responses also apply to most of them.

appraisal of risky activities to assess risk preferences.¹⁰ In this type of scenario, $\mathcal{N}$ is the set of individuals surveyed about their participation in a number of activities $\mathcal{Q}$.

Text analysis. Consider the particular use of language by various individuals or institutions on a given dimension, such as behavioral traits or political attitudes. The purpose is to order a set of linguistic expressions and, as a result, to categorize individuals by their language use. See Laver, Benoit, and Garry (2003) for an application to political texts to extract policy positions.¹¹ Using our terminology, Q_n represents the subset of linguistic expressions used by individual n .

Product Space and Recommendations. Many recommendation systems, such as those for movies or music bands, are based on arranging products and consumers within a characteristic space, with consumers favoring products closer to them.¹² In some cases, objects may be arranged on a single scale, and understanding this can be instrumental in improving recommendation systems. This approach is rooted in classical psychology literature, where tastes are investigated by observing the acceptability or rejection of stimuli that can be ordered. See Pfaffmann (1960) for an early treatment, and consider, as discussed there, products with different salt concentrations as an example of ordered stimuli. Following this line of thought, Q_n represents the set of products liked by individual n .

Wishlisting. The previous example involves recommendations to other users, but one may also consider recommendations to oneself in the form of a wishlist. In this context, the wishlist Q_n represents the set of items for which an individual expresses interest or a positive attitude. See Manzini, Mariotti, and Ülkü (2024) for a sequential model of individual “wishlisting.”

Attention studies. Another case that deviates from the traditional question-response paradigm of surveys involves the use of more implicit expressions of interest. This is common in attentional studies, where a number of individuals $\mathcal{N}$ is offered a number of stimuli $\mathcal{Q}$, and the analyst’s goal is to infer which of these stimuli attract each

¹⁰The authors gathered information on activities such as substance use (e.g., alcohol and marijuana), interpersonal behaviors (e.g., going on a blind date), and sports (e.g., rock climbing). Again, note that these activities do not have a clear ex-ante ranking of riskiness.

¹¹They consider words such as ‘drugs,’ ‘markets,’ ‘nation,’ ‘security,’ or ‘representation,’ and order them in the ideological spectrum. See also the recent use of large language models to position political texts in several unidimensional ideological and policy spaces in Le Mens and Gallego (2024).

¹²The classical Hotelling setting is also based on this simple idea to ultimately model choice behavior.

individual’s attention. Often, objects can be ordered, and individuals pay attention to an interval of stimuli. A classical paper in this field is Thomas (1973), which studies children’s interest in faces drawn with different degrees of detail.¹³

Approval voting. The idea of allowing individuals to pick any number of options has been suggested as a voting mechanism. See Brams and Fishburn (1978) for an early analysis of approval voting. Here, Q_n represents the subset of candidates approved by individual n . Interestingly, Laslier (2009) and Alós-Ferrer and Buckenmaier (2019) have shown that equilibrium behavior in approval voting games may have the upper-contour-set structure analyzed in this paper.¹⁴

Judgment aggregation. There is a set of individuals $\mathcal{N}$ making judgments on a set of logical propositions $\mathcal{Q}$. Dietrich and List (2010) study conditions that facilitate judgment aggregation; one of these conditions involves the case where propositions can be ordered on a line such that the accepted judgments of each individual form an interval.

Networks and homophily. Consider a set of individuals who have directed links to others. The question is whether the network can be explained by ordering individuals such that they form links with those most similar to them. For an application using Twitter data to gather political ideology, see Barberá (2014). We can understand this within our setting by considering $\mathcal{Q} = \mathcal{N}$.

Matching. Let $\mathcal{N}$ be a set of individuals and $\mathcal{Q}$ a set of objects (as in housing problems) or a set of different individuals (as in marriage problems). Individuals submit a set Q_n of acceptable elements. Typically, individuals have a common preference on $\mathcal{Q}$ determined, e.g., by quality. However, in some settings, individuals may have idiosyncratic single-peaked preferences on $\mathcal{Q}$, perhaps given by quasi-linear preferences on quality and price in the housing problem, or by affinity in the marriage problem (see Bade (2019)).

3.2. Revealed information: extremeness. We start by discussing how survey responses reveal the order of questions, which will soon be proven to be the critical

¹³Eye-tracking techniques are used in economics to gather information on attention; see Lahey and Oxley (2016) for a review. Polonio, Di Guida and Coricelli (2015) employ seriation techniques to understand the linkage between saccades and strategies of play within a game.

¹⁴Núñez and Xefteris (2017) study approval voting under single-peaked preferences, which is related to our notion of endorsement-by-proximity. They show that every Nash-implementable welfare optimum can indeed be implemented by means of approval voting mechanisms.

feature for survey rationalizability. First, recall that the upper contour set of a strictly quasi-concave utility function must correspond to an interval on the real line. Then, consider individual n , and suppose we observe $Q_n \cap \{q_1, q_2, q_3\} = \{q_2, q_3\}$. In other words, individual n endorses questions q_2 and q_3 but not question q_1 . Given the interval nature of responses, this observation reveals that q_1 occupies an extreme position within the triple. When this is the case, we write $q_1 \mid \{q_2, q_3\}$ and say that q_1 has been revealed as extreme in the triple $\{q_1, q_2, q_3\}$.

Crucially, any single respondent provides only a fragment of information. The full ordering of questions can only be reconstructed by consistently aggregating these individual-level revelations across the entire population of respondents. For this aggregation to be possible, the revealed information across individuals should not be contradictory. We can then consider the following asymmetry-type of property.

Weak Axiom of Revealed Extremeness (WARE). We say that $\{Q_n\}_{n \in \mathcal{N}}$ satisfies WARE if, for every triple $\{q_1, q_2, q_3\}$ of questions, $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$ imply that $q_3 \mid \{q_1, q_2\}$ cannot hold.

Having observed the revelation of $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$, rationalization requires positioning q_1 and q_2 as extremes within the triple, which in turn necessitates that q_3 is not extreme. It is immediate that WARE is a necessary property for rationalizability. To see this, suppose that $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable and assume that $q_1 \mid \{q_2, q_3\}$ and $q_2 \mid \{q_1, q_3\}$ hold. Then, it must be that $\{\mu_{q_1}, \mu_{q_2}\} = \{\min\{\mu_{q_1}, \mu_{q_2}, \mu_{q_3}\}, \max\{\mu_{q_1}, \mu_{q_2}, \mu_{q_3}\}\}$. Hence, if we observe any (interval) response set containing both q_1 and q_2 , it must also contain q_3 , and WARE is satisfied. However, the following example shows that WARE is not sufficient for rationalization.

Example 1. Consider $\mathcal{N} = \{1, 2, 3\}$, $\mathcal{Q} = \{q_1, q_2, q_3, q_4\}$ and the response sets are $Q_1 = \{q_1, q_2\}$, $Q_2 = \{q_1, q_3\}$ and $Q_3 = \{q_1, q_4\}$. By construction, there are at most two extremeness revelations in every triple of alternatives, and WARE holds. However, the responses are not rationalizable; the information contained in Q_1 requires q_4 to be located as the maximum or minimum in the triple $\{q_1, q_2, q_4\}$. If it is a maximum (respectively, a minimum), the information contained in Q_2 requires q_4 to also be the maximum (respectively, the minimum) in the triple $\{q_1, q_3, q_4\}$ and, as a consequence, the maximum (respectively, the minimum) in the set $\{q_1, q_2, q_3, q_4\}$. However, the same argument can be applied to q_3 by considering response sets Q_1 and Q_3 , and to q_2 by

considering response sets Q_2 and Q_3 . Since the three alternatives cannot be extreme in the set of four alternatives, a contradiction arises and the result follows.

3.3. A characterization of rationalizability. Example 1 demonstrates that the absence of direct contradictions required by WARE is insufficient for rationalizability. We now outline two equivalent approaches for appropriately expanding survey information and characterize survey rationalizability. The first approach involves augmenting the information revealed by the survey through the inclusion of fictitious responses. The second approach extends the extremeness relation to account for all indirect extremeness revelations. While these approaches turn out to be equivalent, they highlight different aspects of the problem, which can be instrumental for addressing various issues, as we demonstrate below in our study of the complexity of testing for rationalizability.

In order to discuss the first approach, we need to consider pairs of overlapping responses, namely:

$$(Q_i, Q_j) \text{ such that } Q_i \cap Q_j \neq \emptyset, Q_i \setminus Q_j \neq \emptyset \text{ and } Q_j \setminus Q_i \neq \emptyset.$$

Recall that, under rationalizability, response sets correspond to intervals of the real line. Hence, the union of two overlapping response sets also corresponds to a (larger) interval.¹⁵ If this larger set of responses does not belong to the original set of responses, it brings new information that can be added to the original survey. We then consider this principle recursively until no more novel unions of overlapping responses can be added. Denote $\mathcal{N}_0 = \mathcal{N}$ and consider a sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$, such that for every $m < M$, the $(m + 1)$ -th D-survey is equal to the m -th D-survey with the addition of a (novel) union of two overlapping sets in the m -th D-survey. Given finiteness, this sequence ends up in a D-survey that admits no further addition, $\{Q_n\}_{n \in \mathcal{N}_M}$, that we call the expanded D-survey of $\{Q_n\}_{n \in \mathcal{N}}$. Importantly, notice that this survey is closed under unions of overlapping responses.¹⁶

As for the second approach, we now define the indirect extremeness revelations derived from the direct extremeness revelations contained in the original D-survey. Let T be a generic ternary relation on a finite set X , and define the following condition,

¹⁵Notice that this statement does not necessarily apply if the responses are not overlapping: if two intervals are disjoint, the union may fail to be an interval. Also, notice that the intersection and difference of overlapping intervals must also be intervals, but our revelation technique does not require including this information.

¹⁶The order in which information is added is irrelevant and the expanded D-survey is unique.

that we call overlapping.¹⁷

$$xT\{y, z_1\} \text{ and } xT\{y, z_2\} \text{ implies } xT\{z_1, z_2\}.$$

Given T , the overlapping closure of T , denoted $\widehat{T}$, is the minimal ternary relation containing T that is closed under overlapping. We then use the overlapping closure of $\mid$ to define a stronger version of WARE.

Strong Axiom of Revealed Extremeness (SARE). We say that $\{Q_n\}_{n \in \mathcal{N}}$ satisfies SARE if, for every triple $\{q_1, q_2, q_3\}$ of questions, $q_1 \widehat{\mid} \{q_2, q_3\}$ and $q_2 \widehat{\mid} \{q_1, q_3\}$ imply that $q_3 \mid \{q_1, q_2\}$ cannot hold.

We are now in a position to state the main theorem of this section, Theorem 1. It shows that D-survey rationalizability is equivalent to the expanded D-survey satisfying WARE, which is itself equivalent to the D-survey satisfying SARE. Importantly, Theorem 1 shows that checking the rationalizability of data can be done in polynomial time, making the problem computationally tractable.

Theorem 1. *The following statements are equivalent:*

- (1) $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable.
- (2) $\{Q_n\}_{n \in \mathcal{N}_M}$ satisfies WARE.
- (3) $\{Q_n\}_{n \in \mathcal{N}}$ satisfies SARE.

Moreover, determining whether $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable is polynomial with a time complexity of $O(Q^4)$.

Theorem 1 demonstrates that survey rationalizability can be understood through a familiar economic approach: (i) include all indirect revelations derivable from the data, and (ii) require these revelations to be asymmetric. Notably, these revelations pertain not to choice exercises but to attitude survey responses. They are unrelated to the preference ordering of a decision-maker and instead concern the positioning of questions/individuals on the corresponding attitude scale. Crucially, these revelations involve a ternary relation, not a binary one.

Statements (2) and (3) of Theorem 1 describe two conceptual views of indirect revelations. Statement (2) involves expanding the survey with fictitious responses and considering the direct revelations of the expanded survey. Instead, statement (3) extends the original survey's direct revelations into an indirect (overlapping closure) relation.

¹⁷The property is a type of transitivity and we use the term overlapping to relate it to the notion of overlapping sets.

The proof of their equivalence uses a recursive argument, showing that (i) at each step, the fictitious response added contains only revelations that could be indirectly derived, and (ii) every possible indirect revelation is contained in a fictitious response added at some step in the sequence.

The equivalence of (2) and (3) with statement (1) is established as follows. First, survey rationalizability can be expressed in terms of the ordinal consecutive-ones property (c1p).¹⁸ In essence, for a given order of questions $\succ$ where c1p holds, one can construct a corresponding strictly quasi-concave utility function and threshold for each agent. This utility function is strictly increasing up to the midpoint of the interval defined by the minimal and maximal positively responded questions. At these two boundary questions, the utility is zero, which is equal to the threshold. The consecutive nature of responses given by c1p ensures this construction is valid for rationalizability.

In order to show that our properties guarantee c1p, we use Theorem 6 in Tucker (1972), which is based on the absence of “asteroidal triples,” a concept introduced by Lekkerkerker and Boland (1962) in their study of interval graphs. In essence, the asymmetry condition contained in WARE corresponds to the simplest possible asteroidal triple, and we need to prove that the asymmetry of indirect revelations rules out any asteroidal triple. This result follows from a recursive argument, with the indirect revelation approach facilitating the analysis.¹⁹

Finally, given the equivalence between (1) and (3), the complexity of checking for rationalizability reduces to the complexity of checking the asymmetry of the overlapping closure of the extremeness relation of the original D-survey. In the proof, we show that this can be done using a version of the classical Floyd-Warshall algorithm that is known to run in polynomial time.

¹⁸See the proof for the formal definitions of c1p and that of asteroidal triple below, within our context.

¹⁹Bergman (2022) offers an alternative potential route. Using our terminology, he works with the smallest collection of response sets that contains the original response data and is closed under unions, intersections, and differences of overlapping sets. Bergman shows that a simple property on this expanded dataset characterizes the existence of a linear order on the collection of questions. For his result to be useful in our setting, one would need to connect his larger expanded dataset to a unique ternary relation and a unique transitivity operation, something that is, at present, far from straightforward, but remains a promising direction for future research. By contrast, we only consider the union of overlapping sets, and our use of Tucker (1972) allows us to create paths between alternatives and connect these paths to our ternary relation and the simple property WARE.

4. POLYTOMOUS AND STOCHASTIC SURVEYS

In this section, we show how to exploit the structure of Theorem 1 when dealing with two related, but informationally richer, problems. The first one covers the Likert scale case in which individuals may use more than two labels to declare their endorsement. The second analyzes the case of dichotomous surveys, in which responses may be probabilistic.

4.1. Polytomous surveys. Given question $q \in \mathcal{Q}$, individual $n \in \mathcal{N}$ is now allowed to express a level of endorsement by using a label from $\mathcal{L} = \{0, 1, \dots, L\}$, where $L > 0$. We write $\mathcal{L}^* = \mathcal{L} \setminus \{0\}$. Higher labels in the collection express stronger endorsement.²⁰ A polytomous survey or, simply, a P-survey is a map $P : \mathcal{N} \times \mathcal{Q} \rightarrow \mathcal{L}$, where $P(n, q) = l$ refers to the case where individual n assigns label l to question q . We consider the following notion of rationalizability, which reduces to the notion used for D-surveys when $L = 1$.

P-survey rationalizability. We say that P is rationalizable whenever there exist $[U_n, \{\tau_n^l\}_{l \in \mathcal{L}^*}]_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave, $\tau_n^1 < \dots < \tau_n^L$, and $\mu_q \in \mathbb{R}$ such that, for every $n \in \mathcal{N}$, $q \in \mathcal{Q}$ and $l \in \mathcal{L}^*$, $P(n, q) = L$ if $U_n(\mu_q) \geq \tau_n^L$, $P(n, q) = l$, $0 < l < L$, if $\tau_n^{l+1} > U_n(\mu_q) \geq \tau_n^l$ and $P(n, q) = 0$ otherwise.

In words, the individual now uses different thresholds to determine her degree of endorsement. These thresholds naturally impose stronger requirements on stronger expressions of endorsement.

We now show that the rationalizability of a P-survey can be obtained building upon Theorem 1, by studying the rationalizability of an associated D-survey. Formally, given the response map P , define the D-survey

$$\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*} \text{ where } Q_{(n,l)}^P = \{q \in \mathcal{Q} : P(n, q) \geq l\}.$$

That is, the vector of responses of individual $n \in \mathcal{N}$ is used to construct L response sets. Each of these response sets contains cumulative information, e.g., response set $Q_{(n,l)}^P$ contains all the questions endorsed by individual n with intensity l or above. Note that, formally, $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ is a D-survey where every (n, l) can be understood as an individual. It is immediate that $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ provides the same information as P does. We need to define the corresponding expanded D-survey, following the same

²⁰For instance, strongly disagree, disagree, neutral, agree, strongly agree.

logic as above, denoted by $\{Q_n^P\}_{n \in \mathcal{N}_M}$. Theorem 2 shows that rationalizability of P is equivalent to this expanded D-survey satisfying WARE.²¹

Theorem 2. *P is rationalizable if and only if $\{Q_n^P\}_{n \in \mathcal{N}_M}$ satisfies WARE.*

4.2. Stochastic responses. We now study the case where survey responses may be stochastic. This can be understood in terms of the variation of responses of the individuals of a given subgroup, like those given by different age or gender groups, or the case in which individual responses express a value in some continuum, such as probabilistic beliefs or expectations, or intensity of support. Stochastic models of responses are also instrumental in dealing with actual data, since they are a way to model and account for noise resulting from, e.g., mistakes due to fatigue or inattention.

Formally, a dichotomous survey with stochastic responses, or S-survey, is a map $S : \mathcal{N} \times \mathcal{Q} \rightarrow [0, 1]$ where $S(n, q)$ describes the probability with which individual $n \in \mathcal{N}$ endorses question $q \in \mathcal{Q}$. Consider the following definition of rationalizability.

S-survey rationalizability. We say that S is rationalizable whenever there exist $\{U_n, F_n\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}$, with $U_n : \mathbb{R} \rightarrow \mathbb{R}$ being strictly quasi-concave, $F_n : \mathbb{R} \rightarrow [0, 1]$ being a cumulative distribution function (CDF) governing the realization of thresholds τ_n , and $\mu_q \in \mathbb{R}$ such that, for every $n \in \mathcal{N}$ and $q \in \mathcal{Q}$, $S(n, q) = F_n(U_n(\mu_q))$.

A question is endorsed whenever the utility of the view it represents is above a threshold, where thresholds are now realized according to a distribution F_n . Hence, endorsement happens probabilistically, and the probability of individual n endorsing question q corresponds to the mass of thresholds below $U_n(\mu_q)$, i.e., $F_n(U_n(\mu_q))$. This is a convenient account of stochastic responses that aligns well with the models studied in this paper and can easily accommodate mistakes at the time of responding. Consider, e.g., an individual who with a large probability considers her true threshold, and with small probability trembles and considers a smaller or a larger threshold. This leads the individual to occasionally endorse questions with an associated utility below her true threshold and to reject questions that are above it.

We now argue that we can rely again on the techniques developed for the analysis of D-surveys. For every $n \in \mathcal{N}$, consider the (at most) Q response sets defined by $Q_{(n,q)}^S = \{q' \in \mathcal{Q} : S(n, q') \geq S(n, q)\}$, and the corresponding D-survey $\{Q_{(n,q)}^S\}_{(n,q) \in \mathcal{N} \times \mathcal{Q}}$. That

²¹For brevity, this and the following result present the characterization based solely on expanded surveys. Nevertheless, the conclusions about overlapping closure and associated complexity remain equally applicable.

is, the information of individual $n \in \mathcal{N}$ appears at most Q times, describing response sets that contain the questions chosen with at least as much probability as a given question. Denote the corresponding expanded D-survey by $\{Q_n^S\}_{n \in \mathcal{N}_M}$. Theorem 3 shows that rationalizability of S is equivalent to this expanded D-survey satisfying WARE.

Theorem 3. *S is rationalizable if and only if $\{Q_n^S\}_{n \in \mathcal{N}_M}$ satisfies WARE.*

5. IDENTIFICATION

In this section, we discuss the identification of the models studied. To present results across different types of surveys, we make the following assumptions: for every individual $n \in \mathcal{N}$: (i) $\emptyset \neq Q_n \neq \mathcal{Q}$, and (ii) $S(n, q) = S(n, q') \Rightarrow P(n, q) = P(n, q') \Rightarrow [q \in Q_n \text{ if and only if } q' \in Q_n]$. That is, (i) implies that we eliminate from the analysis the trivial individuals that endorse everything or nothing, and (ii) connects the response behavior in S-surveys, P-surveys and D-surveys.

We impose a richness condition, according to which for every triple of consecutive questions $\mu_{q_1} > \mu_{q_2} > \mu_{q_3}$, there are two individuals overlapping on this triple.²² In D-surveys this translates into $Q_{n_1} \cap \{q_1, q_2, q_3\} = \{q_1, q_2\}$ and $Q_{n_2} \cap \{q_1, q_2, q_3\} = \{q_2, q_3\}$. For the cases of P-surveys and S-surveys, richness requires that one of the corresponding D-surveys associated to them satisfies the former property.

As discussed above, the models are essentially ordinal, so we need to study the identification of the linear order $\succ$ corresponding to $\{\mu_q\}_{q \in \mathcal{Q}}$. Later, we analyze what we can learn about the location of individuals. Finally, we show that a parametric version of the model allows for cardinal identification results.

5.1. Identification of questions. We start by showing that $\succ$ is fully identified. The reason for this is that, under the richness assumption, the extremeness revelations can be shown to be complete. Having the two extreme questions for every triple, the linear order is fully identified (up to inversion).²³

Proposition 1. *Suppose that a survey is rationalizable with respect to the linear order $\succ$. Under the richness assumption, the indirect revealed relation coincides with $\succ$ (up*

²²As it will become clear, this requirement can be achieved with $Q - 1$ individuals in the expanded survey. If every triple involves different pairs of individuals, then $2(Q - 1)$ individuals would be needed.

²³Since this result holds for all types of surveys, we do not specify the type in the formulation of the result. Also, we simplify the exposition of the proof by showing the key steps using only the D-survey notation.

to inversion). As a result, the survey cannot be rationalizable with respect to any other linear order.

Proposition 1 shows that, under the richness condition, the order of questions is identified by the extremeness revelations. Notice that this idea can also be used even in the absence of the richness condition. In essence, partial identification of the order of questions is obtained: any linear order with an extremeness relation containing the one revealed by the expanded D-survey is plausible.

5.2. Identification of individuals. We now discuss what can be potentially learnt about a specific individual $n \in \mathcal{N}$. As previously discussed, the model is essentially ordinal. Accordingly, here we care about the identification of the location of the ideal view of the individual with respect to $\succ$.²⁴ We show that S-surveys are more informative in this respect than P-surveys, which in turn are more informative than D-surveys. For any individual $n \in \mathcal{N}$, denote by $\underline{q}_n^\alpha$ and $\bar{q}_n^\alpha$, with $\alpha \in \{D, P, S\}$, the minimum and maximum questions (according to μ_q or $\succ$) among: (i) the endorsed ones when $\alpha = D$, (ii) those with the highest strength of endorsement when $\alpha = P$, and (iii) those with the highest probability of endorsement when $\alpha = S$. Then, denote by $\alpha(n)$ the open interval of $\succ$ defined by the question immediately below $\underline{q}_n^\alpha$ and the one immediately above $\bar{q}_n^\alpha$ in $\succ$.²⁵ Importantly, in S-surveys whenever the CDF over the threshold is atomless, these two questions must either coincide or be consecutive, implying that the identification is almost complete.²⁶

Proposition 2. *Under the Richness assumption, the ideal view of individual $n \in \mathcal{N}$ belongs to $S(n) \subseteq P(n) \subseteq D(n)$, and no further identification is possible.*

5.3. Exponential responses. In our non-parametric models, the identification of questions and ideal points is ordinal. We now introduce a stochastic parametric model that illustrates how cardinal identification of the location of questions and ideal points,

²⁴Note that an individual with strictly increasing (decreasing) utility, and hence with no peak on the reals, is indistinguishable in ordinal terms with respect to an individual with a peak above the maximum (below the minimum) of all questions according to $\succ$.

²⁵If one of these two questions does not exist, the position of the ideal view can be seen as unbounded in that specific direction.

²⁶This follows immediately from the fact that, given strict quasi-concavity and the atomlessness of the CDF, only the closest question at each side of the ideal point may come with the highest endorsement probability (up to copies of these questions).

$\{\mu_q\}_{q \in \mathcal{Q}}$ and $\{\mu_n\}_{n \in \mathcal{N}}$ can be achieved. The model postulates that the individual endorsement probability is of the exponential form on the distance between the locations of the question and the ideal point of the individual, $S(n, q) = e^{-\frac{|\mu_n - \mu_q|}{\sigma_n}}$.²⁷ Intuitively, the probability of endorsing a question is 1 when its location coincides with that of the ideal point, decays with the distance between them, and is zero in the limit. It is modulated by a single parameter $\sigma_n > 0$ with a natural interpretation. Smaller values of σ_n accelerate the decay, representing individuals with a lesser tendency to endorse views away from their ideal point. Notice that the exponential endorsement probability could be alternatively presented via our stochastic framework by way of a strictly quasi-concave utility and a random threshold.²⁸

We start by normalizing the location of any two questions, that we denote by 0_q and 1_q , to $\mu_{0_q} = 0$ and $\mu_{1_q} = 1$, respectively.²⁹

Proposition 3. *Let $Q \geq 4$. Under the Richness assumption, in an S -survey with exponential responses, $\{\mu_q\}_{q \in \mathcal{Q} \setminus \{0_q, 1_q\}}$ and $\{\mu_n\}_{n \in \mathcal{N}}$ are cardinally identified.*

6. MULTIDIMENSIONALITY

We now allow for the possibility that questions and individual ideal points belong to a multidimensional space. Given the result below, it suffices to consider the bidimensional case in which $U_n : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\mu_q \in \mathbb{R}^2$. The following result shows that the multidimensional case has no empirical power.

Proposition 4. *In the bidimensional case, every $\{Q_n\}_{n \in \mathcal{N}}$ is rationalizable.*

The intuition behind the result is that, unlike the unidimensional case where only two extreme questions can exist, two-dimensional representations permit an infinite number of extreme questions. Consequently, this imposes no constraints on the responses. Furthermore, the same result holds even if the rationalization concept is strengthened

²⁷The literature in psychology offers various probabilistic models of responses based on the notion of endorsement-by-proximity. See, e.g., Davison (1977). The present model is a convenient version of Andrich (1988) and Hoijtink (1990).

²⁸It suffices to set $U_n(x) = e^{-\frac{|\mu_n - x|}{\sigma_n}}$ and F_n the CDF of the uniform distribution in $[0, 1]$.

²⁹Recall that the inverse of a linear order is equivalent to the linear order for our purposes, so the order of these two alternatives is without loss of generality.

to require strictly convex contour sets.³⁰ Overall, any non-parametric analysis involving more than one dimension becomes trivial.

This result is unsurprising given the existing literature on convex representations of preference profiles. Bogomolnaia and Laslier (2007) demonstrate in their Theorem 16 that every preference profile admits a convex representation in two dimensions. In the current survey setting, we work with a single upper contour set per individual rather than the entire preference profile. As a result, our Proposition 4 can be viewed as a corollary to Theorem 16 of Bogomolnaia and Laslier (2007).³¹

Naturally, one might consider multi-dimensional analysis under versions of the model where utility functions are parametrically constrained. In such cases, not every collection of responses can be rationalized, with, e.g., Euclidean preferences. For example, consider a scenario with 4 individuals and 2^4 alternatives, where each alternative is endorsed by a unique subset of individuals. For these responses to be rationalizable with Euclidean preferences, there would need to exist four circles in the plane forming a Venn diagram. However, this is impossible due to Euler’s relation, which governs the number of faces, edges, and vertices in a planar graph (see, e.g., Ruskey, Savage, and Wagon (2006)). This observation highlights an interesting direction for future research: exploring parametric versions of our otherwise nonparametric framework.

7. DISCUSSION

This paper represents a first attempt in the study of the economic rationality foundations of survey responses, setting the basis for the theoretical and empirical treatment of other important issues. For example, our approach assumes that all individuals share a common understanding of the order of questions. In large-scale surveys, this assumption is likely to be violated, just as standard individual rationality assumptions are often violated in large datasets. This points toward promising future developments in the field, such as methods for assessing goodness-of-fit or quantifying deviations from rationality. One possibility is a rationality index based on the minimal number

³⁰Although the indifference curves used in our construction are polygons formed by a finite number of points on the unit circumference, we can create strictly convex supersets of these polygons that intersect the unit circumference at exactly the same points.

³¹Notice though that the critical unidimensional cases are fundamentally different. Unlike other settings where information on the preference profile is assumed, such as Kalandrakis (2010) or Ballester and Haeringer (2011), the study of survey responses involves only very coarse information about individuals—specifically, a single upper contour set per individual.

of questions that must be ignored for the remaining set to be rationalizable.³² Another is modeling heterogeneity in perceived orderings: for instance, different demographic groups (e.g., by age or gender) may entertain different orders of questions.

Another direction of interest is the combination of surveys with more standard economic data to enhance the information revelation process, in line with the recent suggestions of Caplin (2021), Gerasimou (2021) and Almås, Attanasio, and Jervis (2023).

Also, it is interesting to note that our framework remains valid in the presence of non-truthful responses and, indeed, it provides grounds to empirically study the nature of the potentially existing deviations. Notice first that a respondent trying to appear, say, more moderate would merely present a set of responses with the form of an interval, but shifted towards the moderate part of the scale. Hence, our techniques can still be used in the presence of non-truthful responses, providing correct information on the order of questions, and on the self-presentation of the individuals, who could deviate from their internal views due to conformity, shame, etc. As is common in the survey literature, we could compare survey responses run under different conditions (e.g., degrees of confidentiality or anonymity, different informational environments, or different incentives), or use implicit information (such as response times or measures of attention) to better identify individuals' actual views. The ordered nature of questions obtained in the analysis would facilitate this. Moreover, we can shed light on which variables of the model, e.g., the threshold level or the individual utilities, are affected and how. Alternatively, another interesting approach would entail comparing survey responses with relevant choice data (e.g., risky activities versus lottery choices).

Finally, our framework could provide the basis to study the impact of behavioral considerations on responses. As discussed, our stochastic treatment studied in Section 4.2 provides a flexible tool to incorporate considerations such as mistakes. A possible line of future research would entail explicitly modeling the nature of mistakes. For example, respondents may be affected by fatigue and, as a result, increase or decrease their threshold over the course of the survey.

APPENDIX A. PROOFS

Proof of Theorem 1: (1) $\Rightarrow$ (2). Consider any appropriate sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$ leading to the expanded survey. Given (1), $\{Q_n\}_{n \in \mathcal{N}_0}$

³²This is akin to the Houtman–Maks index in individual choice; see Apesteguia and Ballester (2015) for a study of this and related indices.

is rationalizable. We now claim that if $\{Q_n\}_{n \in \mathcal{N}_m}$ is rationalizable, $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$ is also rationalizable. To see this, suppose that $\{Q_n\}_{n \in \mathcal{N}_m}$ is rationalizable with parameters $[\{U_n, \tau_n\}_{n \in \mathcal{N}_m}, \{\mu_q\}_{q \in \mathcal{Q}}]$. Let $Q_i \cup Q_j$ be the novel response set added to form $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$. This requires Q_i, Q_j to belong to $\{Q_n\}_{n \in \mathcal{N}_m}$ and to be overlapping. To construct the new rationalization, define the location of questions and the individual parameters of any individual other than the novel one as in the rationalization of $\{Q_n\}_{n \in \mathcal{N}_m}$; we then need to define parameters for the novel individual and prove the rationalization of her response set $Q_i \cup Q_j$. To do so, define $a = \min\{\mu_q : q \in Q_i \cup Q_j\}$, $b = \max\{\mu_q : q \in Q_i \cup Q_j\}$ and define the utility of the novel individual to be any strictly increasing utility function up to $\frac{a+b}{2}$ and strictly decreasing after this point, with the utility at a and b being equal to the threshold level. Given the original rationalizability and the overlapping nature of Q_i and Q_j , the interval $[a, b]$ contains exactly questions $Q_i \cup Q_j$, and rationalizability of this response set follows. Given finiteness, this inductive argument guarantees that the expanded D-survey is rationalizable and, as shown in Section 3.2, it must satisfy WARE.

(2) $\Rightarrow$ (1). We use the characterization of c1p of Theorem 6 in Tucker (1972), via the absence of asteroidal triples. In order to be able to use it, we first need to bring the concepts used there to our setting. Formally, c1p requires the existence of a linear order $\succ$ over $\mathcal{Q}$ such that every response set Q_n is formed by consecutive questions in $\succ$. An asteroidal triple is a triple of questions $\{q_1, q_2, q_3\}$ such that for every pair of questions $\{q_i, q_j\}$ in this triple, there exists a sequence of individuals $n_{ij}^1, \dots, n_{ij}^K$ such that: (i) $q_i \in Q_{n_{ij}^1}$ and $q_j \in Q_{n_{ij}^K}$, (ii) for every $k = 1, \dots, K-1$, $Q_{n_{ij}^k} \cap Q_{n_{ij}^{k+1}} \neq \emptyset$, and (iii) the remaining alternative in the triple does not belong to the response set of any individual in this sequence.

We now show that if the expanded D-survey satisfies WARE, the original survey does not contain asteroidal triples. Suppose by contradiction that $\{q_1, q_2, q_3\}$ is an asteroidal triple. Consider first the response sets of n_{ij}^1 and n_{ij}^2 . Given (ii), it must be either that one of these sets contains the other, or that they overlap. In the former case, the larger of these two response sets, denoted $\bar{Q}_{ij}^2$, belongs to the expanded D-survey (as it belongs to the survey), contains q_i but not the remaining alternative from the triple that we denote by q_o , and intersects $Q_{n_{ij}^3}$. In the latter case, since the expanded D-survey is closed under overlapping unions, $\bar{Q}_{ij}^2 = Q_{n_{ij}^1} \cup Q_{n_{ij}^2}$ must belong to the expanded D-survey, contains q_i but not q_o , and intersects $Q_{n_{ij}^3}$. Consider the response sets $\bar{Q}_{ij}^2$ and $Q_{n_{ij}^3}$. Again, it must be either the case that one of them contains the

other or they overlap. With the same reasoning, we can identify $\bar{Q}_{ij}^3$ in the expanded D-survey, containing q_i but not q_o , and intersecting $Q_{n_{ij}}^4$. Proceed recursively until obtaining the response set $\bar{Q}_{n_{ij}}^K$. This set contains q_i and q_j , but not q_o . Since this argument applies to any combination in the triple, we have identified three (fictitious) individuals in the expanded D-survey violating WARE, a contradiction. Now, having proved that the original survey does not contain asteroidal triples, the application of Theorem 6 in Tucker (1972) guarantees that the original survey must satisfy c1p.

We now show that given that the original survey satisfies c1p, it is rationalizable. Given the finiteness of $\mathcal{Q}$, $\succ$ is representable and, hence, there exists a collection of real values $\{\mu_q\}_{q \in \mathcal{Q}}$ such that for every pair of questions $q, q' \in \mathcal{Q}$, it is $\mu_q > \mu_{q'} \Leftrightarrow q \succ q'$. For every $n \in \mathcal{N}$ such that $|Q_n| \geq 1$, consider any utility function $U_n(x)$ that is strictly increasing whenever $x \leq \frac{\min_{q \in Q_n} \mu_q + \max_{q \in Q_n} \mu_q}{2}$, and strictly decreasing otherwise. Moreover, we can select U_n to satisfy $U_n(\min_{q \in Q_n} \mu_q) = U_n(\max_{q \in Q_n} \mu_q) = 0$. This function is strictly quasi-concave. Set $\tau_n = 0$. Since Q_n is formed by consecutive questions in $\succ$, it is immediate that $Q_n = \{q : U_n(\mu_q) \geq \tau_n = 0\}$, proving rationalization. For every $n \in \mathcal{N}$ with $|Q_n| = 0$, we can rationalize responses by considering U_n to be strictly negative and strictly quasi-concave, and setting $\tau_n = 0$.

We now prove that the extremeness relation directly revealed by the expanded D-survey coincides with the overlapping closure of the extremeness relation of the original D-survey, implying that (3) is equivalent to (2). Consider again any appropriate sequence of D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$, $m = 0, 1, \dots, M$, leading to the expanded D-survey. Denote by $|_m$ the extremeness revelations of survey $\{Q_n\}_{n \in \mathcal{N}_m}$. We need to show that $|_M = \widehat{\cdot}$. Consider two successive D-surveys $\{Q_n\}_{n \in \mathcal{N}_m}$ and $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$ in the sequence, and assume that $Q_i \cup Q_j$, with $Q_i, Q_j \in \{Q_n\}_{n \in \mathcal{N}_m}$, is the novel response in $\{Q_n\}_{n \in \mathcal{N}_{m+1}}$. We claim that $|_{m+1}$ is composed of $|_m$ plus every overlapping $q \mid_m \{q', q_1\}, q \mid_m \{q', q_2\}$ in which $q \notin Q_i \cup Q_j$, $q_1 \in Q_i \setminus Q_j$ and $q_2 \in Q_j \setminus Q_i$. First, it is obvious that any such overlapping will be revealed by $Q_i \cup Q_j$ and, hence, present in $|_{m+1}$. Second, consider any revelation $q \mid_{m+1} \{q', q''\}$ that was not present in $|_m$. It must be the case that this revelation is generated by response set $Q_i \cup Q_j$, and for it to be novel, it must be $q \notin Q_i \cup Q_j$, $q' \in Q_i \setminus Q_j$ and $q'' \in Q_j \setminus Q_i$. But in such a case, since Q_i and Q_j are overlapping sets, there must exist $q^* \in Q_i \cap Q_j$, and the D-survey $\{Q_n\}_{n \in \mathcal{N}_m}$ must reveal $q \mid_m \{q^*, q'\}$ (via Q_i) and $q \mid_m \{q^*, q''\}$ (via Q_j), and hence the novel revelation can be seen as an overlapping of two original revelations. Using this recursively, we have shown that the extremeness revelations of the expanded D-survey can

be seen as overlappings of the original extremeness relation and hence $|\equiv|_0 \subseteq |_M \subseteq \widehat{|\equiv|}$. We now claim that $|_M$ is closed under overlappings. To see this, suppose that we observe $q |_M \{q^*, q'\}$ and $q |_M \{q^*, q''\}$. Then, we know that there exists Q_i, Q_j in the expanded D-survey such that $q^* \in Q_i \cap Q_j$ and $q' \in Q_i$ and $q'' \in Q_j$. If $q'' \in Q_i$ or $q' \in Q_j$, we can use the corresponding set to claim that $q |_M \{q', q''\}$. If none is true, then Q_i and Q_j are overlapping sets and, since the expanded D-survey is closed under overlapping unions, $Q_i \cup Q_j$ belongs to the expanded D-survey, and we can use it to claim that $q |_M \{q', q''\}$. Since the overlapping closure is the minimal ternary relation containing $|$ and closed under overlapping, the following must hold $\widehat{|\equiv|} \subseteq |_M$.

Finally, in order to establish the computational complexity of survey rationalizability, consider the graph formed by $Q + Q(Q-1)/2$ nodes corresponding to all questions and pairs of questions in the D-survey. Connect every question q to every pair of questions $\{q', q''\}$, and associate to their edge a weight of 1 when $q | \{q', q''\}$, and a weight of ∞ otherwise. We denote these weights by $[q, \{q', q''\}, 0]$. We then use the Floyd-Warshall algorithm over this (survey) graph, in order to identify the shortest path from q to $\{q', q''\}$, possibly using other nodes as intermediate. Denoting by $[q, \{q', q''\}, l]$ the weight of the shortest path from q to $\{q', q''\}$, possibly involving the use of intermediate questions $\{1, \dots, l\}$ via overlapping, it is immediate that these values satisfy

$$[q, \{q', q''\}, l] = \min\{[q, \{q', q''\}, l-1], [q, \{q', l\}, l-1] + [q, \{q'', l\}, l-1]\}.$$

We can iterate this process until l reaches Q . Note that edges with a finite value at the end of the process must correspond to the overlapping closure of $|$. Using the analogous reasoning to the standard result in which the Floyd-Warshall algorithm computes the transitive closure of a binary relation in polynomial time $\mathcal{O}(Q^3)$, it follows that: (i) computing the overlapping closure of the extremeness ternary relation can be done in polynomial time, and (ii) given the Q^2 order of the nodes of the survey graph (instead of Q in the transitive closure) this computation has polynomial order $\mathcal{O}(Q^4)$. This concludes the proof of the theorem. $\blacksquare$

Proof of Theorem 2: To see the only if part, suppose that P is rationalizable. Then, it must be so for some collection of parameters $[\{U_n, \{\tau_n^l\}_{l \in \mathcal{L}^*}\}_{n \in \mathcal{N}}, \{\mu_q\}_{q \in \mathcal{Q}}]$. Consider the D-survey $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$. By associating to response set $Q_{(n,l)}^P$ the utility function U_n and the threshold τ_n^l , this D-survey must be rationalizable. We can then use the only if part of Theorem 1 and the expanded D-survey $\{Q_n^P\}_{n \in \mathcal{N}_M}$ must satisfy WARE.

Suppose now that $\{Q_n^P\}_{n \in \mathcal{N}_M}$ satisfies WARE. By using the if part of Theorem 1, the D-survey $\{Q_{(n,l)}^P\}_{(n,l) \in \mathcal{N} \times \mathcal{L}^*}$ must satisfy c1p. We now show it is rationalizable. Let $\succ$ be a linear order on $\mathcal{Q}$ that guarantees so and consider any set of values $\{\mu_q\}_{q \in \mathcal{Q}}$ such that $\mu_q > \mu_{q'} \Leftrightarrow q \succ q'$. Given individual $n \in \mathcal{N}$, let $l_n > 0$ be the highest label used by this individual. If this label is used to describe endorsement for at least two questions, define $\mu_n = \frac{\min_{q \in Q_{(n,l_n)}^P} \mu_q + \max_{q \in Q_{(n,l_n)}^P} \mu_q}{2}$ and set $U_n(\mu_n) = l_n$ and $U_n(\min_{q \in Q_{(n,l_n)}^P} \mu_q) = U_n(\max_{q \in Q_{(n,l_n)}^P} \mu_q) = l_n - 1$. All values in the interval $[\min_{q \in Q_{(n,l_n)}^P} \mu_q, \max_{q \in Q_{(n,l_n)}^P} \mu_q]$ are set by considering the piecewise linear function determined by these three values. If the label was used for only one question, set μ_n equal to the μ -value of that question, and $U_n(\mu_n) = l_n - 1$. After this, consider recursively the rest of the labels, starting with $l_n - 1$ down to label 1. For any label l , determine the questions $\min_{q \in Q_{(n,l)}^P} \mu_q$ and $\max_{q \in Q_{(n,l)}^P} \mu_q$. If any of them is different from those previously considered (i.e., whenever these questions have label l exactly), extend the piecewise linear function U_n by forcing it to adopt the value $l - 1$ at the location of these new questions. After label 1, extend the function beyond its extremes with any strictly monotone linear functions. Set thresholds $\tau_n^l = l - 1$. The function U_n is strictly quasi-concave, and since $Q_{(n,l)}^P$ is formed by a set of consecutive questions in $\succ$, it is immediate that questions in $Q_{(n,l)}^P$, and only these questions, have utility above threshold τ_n^l . Rationalization follows. ■

Proof of Theorem 3: The proof is analogous to that of Theorem 2, so we only mention the most significant differences. Notice that from S we can construct a set of at most $Q \times N$ labels, and define the P-survey with the response map P having the property that, for every $n, n' \in \mathcal{N}$ and $q, q' \in \mathcal{Q}$, $P(n, q) \geq P(n', q')$ if and only if $S(n, q) \geq S(n', q')$. This allows us to construct the values μ_q and the functions U_n exactly as in Theorem 2. For every τ_n such that there exists $q \in \mathcal{Q}$ with $U_n(\mu_q) = \tau_n$, set $F_n(\tau_n) = S(n, q)$. Now, for every individual $n \in \mathcal{N}$, a larger threshold corresponds to a question with a larger utility and given the construction and the assumption that $P(n, q) \geq P(n, q')$ if and only if $S(n, q) \geq S(n, q')$, a larger utility corresponds to a larger probability of endorsement. We can then extend these values to a CDF F_n over the reals, and rationalization follows. ■

Proof of Proposition 1: Let the survey be rationalizable. Suppose without loss of generality that it is so for the linear order $Q \succ Q - 1 \succ \dots \succ 1$. We now claim that for every triple of different questions in this sequence satisfying $i > j > k$, the expanded D-survey must reveal $i \mid \{j, k\}$ and $k \mid \{i, j\}$, i.e., that i, k are extreme in the triple

(or, in a terminology that simplifies the exposition, that j must be the intermediate question in the triple). We prove this claim recursively, over the difference $i - k$. The result is trivially true for $i - k = 2$ because in this case the triple must be formed by three consecutive questions and the richness condition guarantees that two individuals exist revealing the intermediateness of j . Suppose now that the result is true up to some value $i - k = t \geq 2$ and consider a triple of questions such that $i - k = t + 1$. Consider question $j^* \neq j$ such that $i > j^* > k$, which must exist because $i - k = t \geq 2$. Let us assume that $j^* > j$ (the other case is dual and omitted). Given the recursive step, we know that $i \mid \{j, j^*\}$ and $j \mid \{i, j^*\}$, i.e., j^* has been revealed intermediate in $\{i, j^*, j\}$. We also know that $k \mid \{j, j^*\}$ and $j^* \mid \{j, k\}$, i.e., j has been revealed intermediate in $\{j^*, j, k\}$. Therefore, there exist four individuals $m_1, m_2, m_3, m_4 \in \mathcal{N}_M$ such that $Q_{m_1} \cap \{i, j^*, j\} = \{i, j^*\}$, $Q_{m_2} \cap \{i, j^*, j\} = \{j^*, j\}$, $Q_{m_3} \cap \{j^*, j, k\} = \{j^*, j\}$, and $Q_{m_4} \cap \{j^*, j, k\} = \{j, k\}$. Suppose first that $i \in Q_{m_3}$. Then, by rationalizability, since $j^* \notin Q_{m_4}$, it must be that $i \notin Q_{m_4}$ and, as a result, individuals m_3 and m_4 reveal that $k \mid \{i, j\}$ and $i \mid \{j, k\}$, as desired. Suppose now that $i \notin Q_{m_3}$. In this case, notice that by rationalizability, $k \notin Q_{m_1}$ (because it does not contain $j > k$) and, by assumption, $k \notin Q_{m_3}$. Since Q_{m_1} and Q_{m_3} are overlapping, the response $Q_{m_1} \cup Q_{m_3}$ belongs to the expanded D-survey and $[Q_{m_1} \cup Q_{m_3}] \cap \{i, j, k\} = \{i, j\}$. Then, responses $[Q_{m_1} \cup Q_{m_3}]$ and Q_{m_4} reveal that $k \mid \{i, j\}$ and $i \mid \{j, k\}$, as desired. All intermediate revelations of $\succ$ have been obtained through the extremeness revelations. Finally, to show uniqueness, consider any other order $\succ'$ different than $\succ$ and its inverse. Hence, there must be three questions $i > j > k$ where the intermediate one is different from j . Without loss of generality, let i be the intermediate question in $\succ'$. But then, we know that there is a response in the expanded D-survey such that $Q_m \cap \{i, j, k\} = \{j, k\}$, and this is not rationalizable for $\succ'$. If the expanded D-survey is not rationalizable with $\succ'$, the proof of Theorem 1 guarantees that the original survey cannot be rationalizable with $\succ'$, concluding the proof. $\blacksquare$

Proof of Proposition 2: Let the survey be rationalizable and $\succ$ be the unique linear order, up to inversion, guaranteeing so. Consider $n \in \mathcal{N}$ and an α -survey. First, we claim that the ideal view must belong to the interval $\alpha(n)$. Suppose by way of contradiction that this is not the case. We analyze the case in which the ideal view occupies a position equal to or above the question immediately above $\bar{q}_n^\alpha$, and omit the other, dual case. In this case, notice that the question $\bar{q}_n^\alpha$ achieves a higher utility than

the question above, contradicting the quasi-concavity of U_n . Second, we claim that $S(n) \subseteq P(n) \subseteq D(n)$. To see this, consider first $q \in S(n)$. By construction, q achieves the highest endorsement probability in $\mathcal{Q}$ and, given our basic assumptions, there exists at least one question q' with strictly less endorsement probability. Question q must belong to the equivalence class of questions with the highest label in the P-survey, and this level must be different than zero. It then follows that $S(n) \subseteq P(n)$. Consider now a question $q \in P(n)$. By construction, q achieves the highest label in $\mathcal{Q}$ and, given our basic assumptions, this label is strictly above zero. Question q must belong to the response set in the D-survey and $P(n) \subseteq D(n)$. Third, we claim that no further identification is possible. We analyze the case of D-surveys, with the other two being analogous. Consider the set Q_n . Modify the selection of the ideal point in the proof of Theorem 1 to occupy any position in $D(n)$. If this position is between $\underline{q}_n^\alpha$ and $\bar{q}_n^\alpha$, set the utility function to be piecewise linear, with value 1 for the ideal view and 0 for these two questions. Otherwise, it must be either above $\bar{q}_n^\alpha$ or below $\underline{q}_n^\alpha$. Consider the first case (the second being omitted as it is analogous). Then, set the utility function to be piecewise linear, with value 1 for the ideal view, 0 for $\underline{q}_n^\alpha$, and -1 for the question immediately above $\underline{q}_n^\alpha$ (if no question exists there, any strictly decreasing function works). This alternative representation also provides rationalizability and completes the proof. ■

Proof of Proposition 3: Consider an S-survey with exponential responses. By Proposition 1, we know that $\succ$ is fully identified, and we set it to be $1_q \succ 0_q$. Given the atomless nature of the exponential model, for every $n \in \mathcal{N}$, the interval $S(n)$ contains either one or two questions. As in the proof of Proposition 2, denote these two questions by $\bar{q}_n^S$ and $\underline{q}_n^S$, and it must be either $\bar{q}_n^S \succ \underline{q}_n^S$ or $\bar{q}_n^S = \underline{q}_n^S \equiv q_n^S$. We now construct a map $h_n : \mathcal{Q} \rightarrow \{0, 1\}$ for each individual such that $h_n(q) = 1$ if and only if $\mu_q \geq \mu_n$ and $h_n(q) = 0$ otherwise. To do so, we consider two cases. If $|S(n)| = 2$, the exponential model of responses guarantees that μ_n must occupy a position strictly between these two questions, and we can assign $h_n(q) = 1$ to $\bar{q}_n^S$ and to every question above this one, according to $\succ$, and zero otherwise. If $|S(n)| = 1$, the map can be defined by $h_n(q) = 1$ whenever $q \succ q_n^S$ and $h_n(q) = 0$ whenever $q_n^S \succ q$. However, the relative location of q_n^S is yet undetermined. To do so, we consider another individual n' for which $h_{n'}(q_n^S)$ is known, and two questions q', q'' for which $h_n(q'), h_n(q''), h_{n'}(q'), h_{n'}(q'')$ are all known.

It is immediate to see that the exponential model requires that

$$\begin{aligned} \frac{(-1)^{h_n(q_n^S)} \log S(n, q_n^S) - (-1)^{h_n(q'')} \log S(n, q'')}{(-1)^{h_n(q')} \log S(n, q') - (-1)^{h_n(q'')} \log S(n, q'')} &= \frac{\mu_{q_n^S} - \mu_{q''}}{\mu_{q'} - \mu_{q''}} = \\ &= \frac{(-1)^{h_{n'}(q_n^S)} \log S(n', q_n^S) - (-1)^{h_{n'}(q'')} \log S(n', q'')}{(-1)^{h_{n'}(q')} \log S(n', q') - (-1)^{h_{n'}(q'')} \log S(n', q'')}. \end{aligned}$$

Hence, $h_n(q_n^S)$ can be determined.³³ Given the complete map h_n , notice now that for every q other than 0_q and 1_q it is

$$\begin{aligned} \frac{(-1)^{h_n(q)} \log S(n, q) - (-1)^{h_n(0_q)} \log S(n, 0_q)}{(-1)^{h_n(1_q)} \log S(n, 1_q) - (-1)^{h_n(0_q)} \log S(n, 0_q)} &= \mu_q, \\ \frac{-(-1)^{h_n(0_q)} \log S(n, 0_q)}{(-1)^{h_n(1_q)} \log S(n, 1_q) - (-1)^{h_n(0_q)} \log S(n, 0_q)} &= \mu_n. \end{aligned}$$

All locations are fully determined, concluding the proof. ■

Proof of Proposition 4: Consider the collection of D-survey responses $\{Q_n\}_{n \in \mathcal{N}}$. We construct a rationalization of these responses. Locate all questions at different positions on the circumference of the unit circle centered at the origin. Consider Q_n . If $|Q_n| = 0$, rationalization is trivial by considering Euclidean preferences centered at the origin, with utility 0 at the origin and $\tau_n = 0$. If $|Q_n| = 1$, rationalization is trivial, by considering Euclidean preferences centered at the only question endorsed, with utility 0 at that point, and $\tau_n = 0$. If $|Q_n| \geq 3$, consider the convex hull of the locations of the questions in Q_n ; denote this set by $CH(Q_n)$. Given that all questions in $\mathcal{Q}$ are at distinct points on the unit circle, it is evident that $q \in CH(Q_n) \cap \mathcal{Q}$ if and only if $q \in Q_n$; every question in $\mathcal{Q} \setminus Q_n$ lies outside this convex hull. We construct the utility function U_n as follows. Select any point in the interior of $CH(Q_n)$ and denote it by μ_n . Then,

- $U_n(\mu_n) = 0$.
- For every y in the boundary of $CH(Q_n)$, set $U_n(y_1, y_2) = -1$.
- For every x outside $CH(Q_n)$, let y be the unique point in the boundary of $CH(Q_n)$ such that $y = \lambda x + (1 - \lambda)\mu_n$. We can then set $U_n(x) = \frac{-1}{\lambda}$.
- For every x in the interior of $CH(Q_n)$, let y be the unique point in the boundary of $CH(Q_n)$ such that $x = \lambda y + (1 - \lambda)\mu_n$. We can then set $U_n(x) = -\lambda$.

³³Notice that in the extreme case in which $S(n, q_n^S) = 1$, it must be $\mu_n = \mu_{q_n^S}$ and $h_n(q_n^S)$ can be freely assigned.

By construction, all indifference curves have the same shape (expansions or contractions of the polygon created by the set of positively endorsed questions by the individual), with utility normalized to 0 and -1 for the ideal point and the polygon indifference curve. This is clearly a quasi-concave utility function which, given the finiteness of the survey, can be immediately transformed into a strictly quasi-concave one. By setting $\tau_n = -1$, this rationalizes the responses of the individual.

If $|Q_n| = 2$, select a third point in the circumference that does not correspond to the location of any question, and consider the triangle formed by these three points. Then, proceed to construct the utility function and the threshold as per the above reasoning. This again rationalizes the set of responses. $\square$

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